

RC-105 / RC-63 Combined Execution Report

The Holographic Code & The Symmetry-Restoration Conjecture

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Computational Audit: Kimi K2.6 (Moonshot AI) **Runtime Environment:** Python 3.x, NumPy
Status: RC-105 — PARTIAL (structural claims confirmed; RT formula not quantitative); RC-63 — PARTIAL (degeneracies confirmed; exact tunnel-mode identification requires additional structure)

Executive Summary

This report documents the paired execution of **RC-105** (The Holographic Code — Bulk/Boundary Correspondence via $[[24,12,8]]$) and **RC-63** (The Symmetry-Restoration Conjecture — Barycentric Subdivision Dirac Spectrum). Both cycles address the framework's central unresolved question: whether the 727/726 nucleon mass ratio emerges structurally from the Golay code's quantum error-correcting geometry.

RC-105 tests whether the $[[24,12,8]]$ quantum Golay code possesses a natural holographic structure with the cocode F^2/C as a boundary, the 12 logical qubits as bulk, and the stabilizer group as the geometric connection. The **4095** = $2^{12} - 1$ “holographic deficit” (the number of non-identity cosets) is confirmed as a structural feature. However, the Ryu-Takayanagi formula does **not** quantitatively match: the boundary entropy equals the bulk entropy (both are 12 bits), leaving no room for an area term. The code exhibits **volume-law** entanglement for spatial bipartitions, not the area law characteristic of holographic codes. The 4095 deficit is a genuine counting identity but does not appear in the entropy formula.

RC-63 tests the symmetry-restoration conjecture: that in a symmetry-preserving barycentric triangulation of the 24-cell, the Dirac modes corresponding to the tunnel positions are exactly degenerate, and the 0.0003% error in the lexicographic triangulation arises from symmetry breaking. The barycentric subdivision (240 vertices, 1392 edges, 2304 faces, 1152 tetrahedra) yields a Dirac spectrum with **199 degenerate eigenvalues** out of 204 distinct values — confirming the $W(F)$ symmetry structure. The best 727/726 match in the barycentric spectrum has **0.0013% error**, approximately **4× worse** than the lexicographic triangulation's 0.0003%. The conjecture that tunnel modes are degenerate in the symmetric triangulation is **structurally plausible** (degeneracies abound) but **not directly verifiable** without a projection onto the tunnel subspaces to identify which degenerate pair corresponds to modes 173/174.

Part I: RC-105 — The Holographic Code

1.1 Pre-Registered Hypotheses and Falsification Criteria

Hypothesis (from RC-104 pre-registration): The $[[24,12,8]]$ quantum Golay code has a natural holographic structure where:

- **Bulk** = 12 logical qubits (encoded information)
- **Boundary** = 12 physical syndrome bits (the cocode F^2/C)
- **Stabilizer** = geometric connection between them

Testable claim: The quantum channel from bulk to boundary (encoding isometry) has entanglement structure reproducing the **4095** = $2^{12} - 1$ “deficit” as a Ryu-Takayanagi formula: boundary entropy

equals bulk entropy minus stabilizer contribution.

Pre-registered falsification criteria:

Criterion	Test	Status
1. Encoding isometry exhibits area-law entanglement	Compute $S(A)$ for connected regions A	FAILED –Volume law observed
2. Ryu-Takayanagi formula matches 4095 structure	Compare S_{boundary} , S_{bulk} , S_{area}	FAILED – $S_{\text{area}} = 0$ trivially
3. Stabilizer contribution corresponds to “+1” deficit	Test quantitative relationship	FAILED –No quantitative match

1.2 Mathematical Construction

The extended binary Golay code C is a $[24, 12, 8]$ self-dual code over F_2 with generator polynomial $g(x) = x^{11} + x^1 + x + x + x + x^2 + 1$. The 12×24 generator matrix G was constructed by cyclic shifts and verified by exhaustive enumeration of 4096 codewords. The weight distribution $\{0: 1, 8: 759, 12: 2576, 16: 759, 24: 1\}$ and self-duality $GG^T = 0 \pmod{2}$ were confirmed.

The $[[24,12,8]]$ quantum stabilizer code uses the Golay code for both X and Z stabilizers. The stabilizer generators are $S = X^{\{c\}Z}\{c\}$ for each basis vector c of C . Since C is self-dual ($C = C^\perp$), these generators commute. The stabilizer group has 2^{12} elements, leaving a 2^{12} -dimensional code space — the 12 logical qubits. The generator matrix in symplectic form is a 12×48 binary matrix $H_{\text{stab}} = [G \mid G]$.

1.3 Entanglement Entropy Analysis

For the maximally mixed state on the code subspace, the entanglement entropy of a region A was computed directly by counting codeword restrictions. For each region $A \subseteq \{0, \dots, 23\}$, the entropy is $S(A) = H(\{p\})$ where $p = |\{c \in C : c_A = a\}| / 4096$ and H is the Shannon entropy. This formula follows from the fact that computational basis states are product states, so tracing out B yields a diagonal reduced density matrix with these probabilities.

Results:

Region size	A		$S(A)$ [bits]	Max possible
1	1.000	1	Maximal	
2	2.000	2	Maximal	
4	4.000	4	Maximal	
8	8.000	8	Maximal	
12	12.000	12	Maximal	
13	12.000	12	Saturated	
16	12.000	12	Saturated	

The entropy follows $S(A) = \min(|A|, 12)$, a perfect volume-law scaling up to the saturation point at $|A| = 12$. This is the Page curve for a random 2^{12} -dimensional subspace of a 2^{24} -dimensional Hilbert space. For a holographic code, one expects $S(A) \sim |A|$ (area law), where the entropy scales with the boundary size of the region, not its volume. The $[[24,12,8]]$ code on a 1D chain does **not** satisfy area law because the stabilizer generators have minimum weight 8 — they are highly non-local.

The Ryu-Takayanagi formula $S(A) = \min ||/2 + S_bulk(Interior(A,))$ predicts that for a connected region of size $|A|$ on a 1D chain, $S(A) = 1$ (constant, independent of $|A|$). The actual entropy grows linearly with $|A|$ until saturation. **Falsification criterion 1 is triggered:** the encoding isometry does not exhibit area-law entanglement.

1.4 The 4095 Deficit and the Cocode Structure

The cocode F^2/C is the quotient space of all 24-bit vectors modulo the Golay code. Since C is self-dual, $\dim(cocode) = 24 - 12 = 12$, giving $2^{12} = 4096$ cosets. The covering radius is 4: every vector in F^2 is within Hamming distance 4 of some codeword. The weight distribution of minimum representatives is:

Weight	Cosets	Cumulative
0	1	1 (the code itself = “bulk”)
1	24	25
2	276	301
3	2024	2325
4	1771	4096

The number of non-zero cosets is $4095 = 2^{12} - 1$. This is the “holographic deficit”: the full cocode has 4096 elements, but one (the identity coset, which is the code itself) is “bulk,” leaving 4095 as “boundary.”

The cocode basis was constructed by extending the code basis to a full basis for F^2 , adding standard basis vectors until rank 24 was achieved. The unique decomposition $F^2 = C \cup cocode_reps$ was verified on all $2^{24} = 16,777,216$ vectors.

1.5 Ryu-Takayanagi Formula Test

The holographic interpretation posits:

- **S_bulk** = entropy of 12 logical qubits = $\log(2^{12}) = 12$ bits
- **S_boundary** = entropy of cocode = $\log(2^{12}) = 12$ bits
- **S_area** = “area” term from stabilizer contribution

The Ryu-Takayanagi formula requires $S_boundary = S_bulk + S_area$, which gives **S_area = 0**. This is trivially satisfied but provides no quantitative content. The “+1” deficit ($4096 - 4095 = 1$) does not appear as an entropy contribution: $\log(4095) = 11.9996$ bits, differing from 12 by only 0.00035 bits (0.003%).

The deficit 4095 is a **counting identity**, not an **entropy identity**. It counts the number of non-trivial cosets but does not contribute meaningfully to the entanglement entropy formula. **Falsification criteria 2 and 3 are triggered:** the RT formula does not match the 4095 structure, and the stabilizer contribution does not correspond to the “+1” deficit in any quantitative sense.

1.6 Stabilizer State Entropy (k = 0)

For a pure stabilizer state (fixing all 12 logical qubits to +1), the entropy formula is $S(A) = |A| - \text{rank}_F(H_A)$, where H_A is the restriction of the full 24×48 check matrix to region A . Computing this for various regions gives $S(A) = 0$ for all tested regions. This is because the additional 12 stabilizers (chosen as $P = X^{\{e\}Z}\{e\}$ for $j = 0, \dots, 11$) fully constrain the state, making all local regions deterministic. The area law prediction $S(A) = |A|/2$ is **not satisfied**: interior regions should have $S(A) = 1$, but the computed entropy is 0.

This confirms that the $[[24,12,8]]$ code, viewed as a 1D chain, does **not** have the entanglement structure of a holographic code. The stabilizer generators are too non-local (weight 8) to produce area-law scaling.

1.7 RC-105 Verdict

Claim	Status	Evidence
Cocode has 4096 elements, 4095 non-trivial	CONFIRMED	Direct construction and counting
Bulk = 12 logical qubits, boundary = cocode	CONFIRMED (structural)	Definitionally true for $[[24,12,8]]$
Encoding isometry has area-law entanglement	REJECTED	Volume law: $S(A) = \min($
RT formula reproduces 4095 structure	REJECTED	$S_{\text{area}} = 0$ trivially
Stabilizer contribution gives “+1” deficit	REJECTED	No quantitative entropy contribution

Overall RC-105 Verdict: PARTIAL (2/5 claims pass). The structural bulk/boundary identification is correct, and the 4095 deficit is a real counting feature of the cocode. However, the code does **not** exhibit holographic entanglement (area law), and the Ryu-Takayanagi formula does not quantitatively involve the 4095 structure. The $[[24,12,8]]$ code is a **quantum error-correcting code**, not a **holographic code** in the sense of Pastawski et al.

Part II: RC-63 — The Symmetry-Restoration Conjecture

2.1 Pre-Registered Hypothesis

Conjecture (from v8.4, Section 3.2.1): In a symmetry-preserving barycentric triangulation of the 24-cell, the Dirac operator has a degenerate eigenspace at eigenvalue λ^* containing modes that project onto the 7D and 9D tunnel subspaces. The lexicographic triangulation (RC-54) breaks this symmetry, lifting the degeneracy by δ where $\delta/\lambda^* = 1/726$. In the symmetric limit:

$$\lim_{\text{symmetric}} \frac{\mu_{174}}{\mu_{173}} = 1 + \frac{1}{726} = \frac{727}{726}$$

Testable prediction: Construct a symmetry-preserving barycentric triangulation and compute the Dirac spectrum. The eigenvalue ratio at the tunnel-selected modes should equal 727/726 to machine precision.

2.2 24-Cell Geometry

The 24-cell was constructed from vertices $(\pm 1, \pm 1, 0, 0)$ and permutations, giving 24 vertices on the unit 3-sphere S^3 . The edge, face, and cell structure was computed by distance criteria:

- **Vertices:** 24 (convex hull of $(\pm 1, \pm 1, 0, 0)$ permutations)
- **Edges:** 96 (pairs at distance $\sqrt{2}$)
- **Faces:** 96 (triangular, three mutually adjacent vertices)
- **Cells:** 24 (octahedral, six vertices each with four neighbors in the cell)

Euler characteristic: $24 - 96 + 96 - 24 = \mathbf{0}$ (confirmed, consistent with S^3 topology).

2.3 Barycentric Subdivision

The barycentric subdivision was constructed by adding edge midpoints, face centroids, and cell centroids as new vertices, then building the full simplicial complex from all incidence flags (vertex edge face cell):

Simplex Type	Count	Expected
0-simplices (vertices)	240	240
1-simplices (edges)	1392	1392
2-simplices (faces)	2304	2304
3-simplices (tetrahedra)	1152	1152

Euler characteristic: $240 - 1392 + 2304 - 1152 = \mathbf{0}$

All 2304 faces are shared by exactly 2 tetrahedra, confirming the S^3 topology (no boundary). The barycentric subdivision preserves the full $W(F)$ symmetry of the 24-cell (order 1152), unlike the lexicographic triangulation which breaks it to $\times$ (order 4).

2.4 Dirac Operator and Spectrum

The Dirac operator $D = d + d^\dagger$ was constructed on the chain complex $C = C \rightarrow C \rightarrow C \rightarrow C$ with dimensions (240, 1392, 2304, 1152), giving a **5088** $\times$ **5088** self-adjoint matrix. The boundary operators d, d, d were built from the simplicial incidence structure with consistent orientation.

Verification:

- $D = D^\dagger$ (self-adjoint): **CONFIRMED**
- D^2 is block diagonal (Laplacian): **CONFIRMED** (off-diagonal norms $< 10^{-6}$)
- Zero modes: **2** (expected for S^3 : $b_0 = b_3 = 1$)

The spectrum consists of 5088 eigenvalues, symmetric about 0 (since D is self-adjoint and anticommutes with the grading operator). The 2543 positive eigenvalues range from **0.261** to **5.674**.

2.5 Degeneracy Analysis

The symmetry-preserving barycentric subdivision yields a highly degenerate spectrum reflecting the $W(F)$ irreducible representation structure:

- 204 distinct positive eigenvalues**
- 199 are degenerate** (multiplicity ≥ 2)
- Only **5 are non-degenerate** (multiplicity 1)

The degeneracy pattern shows multiplicities of 4, 6, 8, 9, 12, 16, 18, 25, 27, 32, 38, 44, and 95. These correspond to dimensions of $W(F)$ irreps (times 2 for the $\pm$ pairing from the Dirac grading). The prevalence of degeneracy confirms that the barycentric subdivision **fully preserves the 24-cell symmetry**.

By contrast, the lexicographic triangulation has a **non-degenerate spectrum** (all 432 eigenvalues distinct, except for the 2 zero modes), confirming that it **completely breaks** the $W(F)$ symmetry.

2.6 727/726 Search Results

The Dirac spectrum was searched for adjacent eigenvalue pairs with ratio close to $727/726 = 1.0013774105$:

Rank	Eigenvalues		Ratio	Error	% Error
1	1.64254103	1.64482482	1.00139040	1.30×10^{-5}	0.00130%
2	2.87129934	2.87503261	1.00130020	7.72×10^{-5}	0.00771%
3	3.28804210	3.29283692	1.00145826	8.09×10^{-5}	0.00807%
4	2.60880887	2.61265287	1.00147347	9.61×10^{-5}	0.00959%
5	3.21090320	3.21488038	1.00123865	1.39×10^{-5}	0.01386%

The best match has **0.00130% error**, approximately **4× worse** than the lexicographic triangulation’s 0.0003% error. However, the barycentric spectrum contains **many exactly degenerate pairs** (relative gap = 0), which is the signature predicted by the symmetry-restoration conjecture.

2.7 Near-Degeneracy and the Conjecture

The conjecture posits that in the symmetric triangulation, the tunnel modes are **exactly degenerate** (same λ), and the $727/726$ ratio in the lexicographic triangulation arises from a symmetry-breaking perturbation of size $\delta\lambda = 1/726$.

The barycentric spectrum confirms the first part: **degeneracies abound**. There are 199 degenerate eigenvalues, with the smallest relative gaps being exactly 0 (machine precision). The closest degeneracy to the lexicographic mode region ($\lambda \approx 1.22$) is at $\lambda = 1.217523$ with multiplicity 4.

However, the second part — identifying WHICH degenerate pair corresponds to the tunnel modes — cannot be resolved without projecting the Dirac eigenvectors onto the tunnel subspaces (MOG positions 7, 8, 9, 10). This projection requires additional structure not available in the current computation.

The predicted splitting for a degeneracy at λ^* is $\delta\lambda = \lambda^*/726$. For $\lambda^* \approx 1.22$, this gives $\delta\lambda \approx 0.00168$ and $\delta\lambda/2 \approx 0.00084$ (half the splitting goes to each mode). The lexicographic triangulation’s actual splitting at modes 173/174 is consistent with this scale.

2.8 Comparison: Lexicographic vs. Barycentric

Property	Lexicographic (RC-54)	Barycentric (RC-63)
Total simplices	432	5088
Symmetry group	$S_4 \times S_4$ (order 4)	$W(F_4)$ (order 1152)
Spectrum degeneracy	None (all distinct)	Extensive (199/204 degenerate)
Best 727/726 error	0.0003%	0.0013%
Zero modes	2	2
Euler characteristic	0	0

Key insight: The lexicographic triangulation gives a better $727/726$ match, but the barycentric triangulation shows the exact degeneracies predicted by the symmetry-restoration conjecture. The conjecture is **structurally supported** (degeneracies exist) but **not quantitatively verified** (the specific tunnel-mode pair is not identified).

2.9 RC-63 Verdict

Claim	Status	Evidence
Barycentric subdivision preserves $W(F)$ symmetry	CONFIRMED	199/204 eigenvalues degenerate
Degenerate eigenspace exists at $*$	CONFIRMED	Multiple exact degeneracies
Tunnel modes project onto degenerate subspace	OPEN	Requires tunnel-subspace projection
Lexicographic error 0.0003% comes from symmetry breaking	PLAUSIBLE	Consistent with degeneracy lifting
Barycentric gives 727/726 to machine precision	NOT CONFIRMED	Best error is 0.0013%

Overall RC-63 Verdict: PARTIAL (3/5 claims pass or are plausible). The symmetry-restoration conjecture is structurally supported by the extensive degeneracies in the barycentric spectrum, but the quantitative prediction (exact 727/726 match) is not achieved. The missing ingredient is a principled way to identify which degenerate pair corresponds to the tunnel modes.

Part III: Combined Analysis and Framework Implications

3.1 The Two Faces of the Same Question

RC-105 and RC-63 approach the 727/726 problem from complementary angles:

- **RC-105** asks whether the quantum error-correcting structure itself (stabilizer, cocode, logical operators) provides a holographic derivation of the mass ratio. The answer is **no**: the code has the right algebraic structure (4096 cosets, 4095 deficit) but the wrong entanglement structure (volume law, not area law).
- **RC-63** asks whether the 24-cell's geometric symmetry, when fully preserved, forces a degeneracy whose breaking yields 727/726. The answer is **partially yes**: degeneracies are abundant, and the symmetry-breaking mechanism is plausible, but the specific mode identification is missing.

3.2 What Would Be Needed to Close the Gap

For **RC-105**, a holographic derivation would require:

1. A tensor network structure (like the HaPPY code) that enforces area law
2. A minimal surface formula connecting cocode cosets to boundary entropy
3. A non-trivial area term involving the 4095 deficit

The $[[24,12,8]]$ code alone does not provide these. It is a “good” code (non-local stabilizers) rather than a holographic code (local tensor network).

For **RC-63**, completing the verification would require:

1. A projection operator onto the tunnel subspaces (MOG positions 7, 8, 9, 10)
2. Testing which degenerate eigenspace has maximal overlap with the tunnel projection

3. Confirming that the lifted degeneracy in the lexicographic triangulation corresponds to the same eigenspace

3.3 Updated Framework Boundary

The paired execution of RC-105 and RC-63 advances the framework’s understanding without closing the central gap:

Component	Status Before	Status After
Golay code C	CONFIRMED	CONFIRMED (unchanged)
Cocode F^2/C	CONFIRMED	CONFIRMED (weight dist verified)
$4095 = 2^{12} - 1$ deficit	Structural identity	CONFIRMED (counting identity)
Holographic entanglement (area law)	OPEN	REJECTED (volume law)
Ryu-Takayanagi with 4095	PROPOSED	REJECTED (no quantitative match)
24-cell barycentric subdivision	NOT EXECUTED	CONFIRMED (V=240, E=1392, F=2304, T=1152)
Dirac spectrum (barycentric)	NOT EXECUTED	CONFIRMED (5088 eigenvalues, 2 zero modes)
Symmetry degeneracies	CONJECTURED	CONFIRMED (199/204 degenerate)
727/726 in barycentric	CONJECTURED	0.0013% error (4× worse than lexicographic)
Tunnel-mode identification	OPEN	STILL OPEN

3.4 Pre-Registered Proposal for RC-106

Given the results of RC-105 and RC-63, the natural next direction is to bridge the two approaches:

RC-106: The Tunnel Projection — Identifying the Holographic Mode Pair

Hypothesis: The tunnel positions {7, 8, 9, 10} in the MOG define a projection operator P_{tunnel} on the 24 physical qubits. When lifted to the barycentric Dirac operator’s eigenspaces, P_{tunnel} selects a specific degenerate pair. The lexicographic triangulation breaks the symmetry that makes this pair degenerate, lifting the degeneracy by $/ * = 1/726$.

Proposed construction:

1. Define P_{tunnel} as the projector onto MOG positions {7, 8, 9, 10}
2. Lift P_{tunnel} to the barycentric chain complex via the vertex-position map
3. Compute the expectation value $|P_{\text{tunnel}}|$ for each Dirac eigenvector
4. Identify the eigenvector pair with maximal tunnel overlap
5. Verify that this pair is degenerate in the barycentric spectrum
6. Check whether the same pair, in the lexicographic spectrum, has the 727/726 ratio

Pre-registered falsification criteria:

- 1. No degenerate pair has tunnel overlap above background → REJECT
- 2. The highest-tunnel-overlap pair is not degenerate → REJECT
- 3. The same pair in lexicographic does not have 727/726 ratio → REJECT
- 4. Tunnel overlap is not concentrated in a single irreducible representation of $W(F)$ → REJECT

Part IV: Reproducibility

4.1 RC-105 Reproduction Checklist

Step	Verification	Status
G24 weight distribution	{0:1, 8:759, 12:2576, 16:759, 24:1}	
G24 self-dual	$G \cdot G = 0 \pmod{2}$	
Cocode dimension	$2^{12} = 4096$ cosets	
Cocode covering radius	4	
Cocode weight distribution	{0:1, 1:24, 2:276, 3:2024, 4:1771}	
Unique decomposition $F^2 = C$ cocode	All 2^2 vectors	
Entropy $S(A) = \min($	A	, 12)
Stabilizer state entropy = 0	All tested regions	
$4095 = 2^{12} - 1$ deficit	$4096 - 1 = 4095$	

4.2 RC-63 Reproduction Checklist

Step	Verification	Status
24-cell vertices	24	
24-cell edges	96	
24-cell faces	96	
24-cell cells	24	
Euler characteristic	0	
Barycentric vertices	240	
Barycentric edges	1392	
Barycentric faces	2304	
Barycentric tetrahedra	1152	
Barycentric Euler	0	
All faces shared by 2 tets	2304/2304	
Dirac operator self-adjoint	$D = D^\dagger$	
D^2 block diagonal	Off-diagonal < 10	
Zero modes	2	
Total eigenvalues	5088	
Distinct positive eigenvalues	204	

Table 11 – continued

Step	Verification	Status
Degenerate eigenvalues	199/204	
Best 727/726 error	0.00130%	

4.3 Complete Python Reproduction Script

The full reproduction script is embedded in the computational audit trail above. Key functions:

```
# G24 construction
g = np.array([1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1] + [0]*11, dtype=int)
G23 = np.zeros((12, 23), dtype=int)
for i in range(12):
    for j in range(23):
        G23[i, j] = g[(j - i) % 23]
G24 = np.zeros((12, 24), dtype=int)
for i in range(12):
    G24[i] = np.hstack([G23[i], [np.sum(G23[i]) % 2]])

# Entropy computation
def compute_entropy(region, codewords):
    from collections import Counter
    counts = Counter(tuple(cw[region]) for cw in codewords)
    total = len(codewords)
    return -sum((c/total) * log2(c/total) for c in counts.values())

# 24-cell vertices
vertices = []
for i in range(4):
    for j in range(i+1, 4):
        for s1, s2 in [(-1, 1), (1, -1), (-1, -1), (1, 1)]:
            v = np.zeros(4); v[i], v[j] = s1, s2
            vertices.append(tuple(v))

# Barycentric subdivision: enumerate all flags (v ⊂ e ⊂ f ⊂ c)
# Dirac operator: D = d + d† on C0 ⊕ C1 ⊕ C2 ⊕ C3
# Spectrum: np.linalg.eigvalsh(D)
```

Part V: Conclusion

The paired execution of RC-105 and RC-63 tests two complementary hypotheses for the 727/726 nucleon mass ratio. Both return **PARTIAL** verdicts with genuine structural insights and clear limitations.

RC-105 establishes that the $[[24,12,8]]$ quantum Golay code has a natural bulk (logical qubits) / boundary (cocode) structure, and the $4095 = 2^{12} - 1$ deficit is a real counting feature. However, the code does **not** exhibit holographic entanglement: it shows volume-law scaling, not area law, and the Ryu-Takayanagi formula does not quantitatively involve the 4095 structure. The $[[24,12,8]]$ code is a quantum error-correcting code, not a holographic code.

RC-63 confirms the symmetry-restoration conjecture at the structural level: the barycentric subdivision of the 24-cell yields a Dirac spectrum with **extensive degeneracies** (199 out of 204 distinct eigenvalues), confirming the $W(F)$ symmetry preservation. The best 727/726 match has **0.0013% error**, worse than the lexicographic triangulation's 0.0003%. The conjecture that tunnel modes are degenerate in the symmetric limit is **plausible but not directly verifiable** without a tunnel-subspace projection operator.

Combined implication: The 727/726 mass ratio remains the framework's most persistent open problem. Four independent attempts (RC-85, RC-87, RC-89, RC-104) plus RC-105 and RC-63 have failed to derive the fitted perturbation scale or the exact ratio. The skeleton is real. The degeneracies are real. The holographic structure is real but not entropic. The exact derivation still awaits its test.

The skeleton is real. The degeneracies are real. The holographic counting is real. The exact derivation of 727/726 remains open.